About Shafarevich–Tate Group of Algebraic Tori

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1.1 Algebraic Tori

1.1.1 Definition. *A group scheme T over a field k is called a* ***torus*** *if*

$$T \underset{k}{\otimes} \bar{k} \simeq \mathbb{G}_m^n,$$

where n is the dimension of T, and $\bar{k}$ is an algebraic closure of k.

Tori over k can be described as follows:

1.1.2 Theorem. *The functor of taking the character module of an algebraic group defines a duality between the following categories:*

$$\begin{array}{c} \textit{group } k\textit{-schemes} \\ \textit{of multiplicative type} \\ \textit{of finite type} \\ \textit{split over } L \end{array} \simeq^{\mathrm{op}} \begin{array}{c} \textit{Finitely generated} \\ \mathrm{Gal}(L/k)\textit{-modules} \end{array}$$

$$T \longmapsto \hat{T} := \mathrm{Hom}_k\big(T_{\bar{k}}, \mathbb{G}_{m,k}\big)$$

$$\hat{T} \longmapsto T, \qquad T(A) = \mathrm{Hom}\big(\hat{T}, (A \otimes_k L)^\times\big).$$

1.2 Norm Tori

Let us fix the notation: let L/k be a Galois extension of global fields with Galois group G, and let $I_G = \mathrm{Ker}(\mathbb{Z}[G] \xrightarrow{\varepsilon} \mathbb{Z})$ be the augmentation ideal.

1.2.1 Definition. *The torus corresponding to the module $I_G^* := \mathrm{Hom}_{\mathbb{Z}[G]}(I_G, \mathbb{Z})$ (see 1.1.2) is called the* ***norm torus*** *and is denoted by $R^1_{L/k}(\mathbb{G}_m)$.*

1.2.2 Proposition. *For the Weil restriction there exists a natural homomorphism*

$$\mathrm{N} : R_{L/k}(\mathbb{G}_m) \to \mathbb{G}_m,$$

called the norm homomorphism. The kernel of this homomorphism is isomorphic to the norm torus defined above:

$$\mathrm{Ker}(\mathrm{N}) \simeq R^1_{L/k}(\mathbb{G}_m).$$

1.3 The Shafarevich–Tate Group

Let T denote a k-torus split over L.

1.3.1 Definition. *The **Shafarevich–Tate group** of the torus T is defined as the kernel of the homomorphism*

$$Ш(T) = \mathrm{Ker}\Big(H^1(G, T(L)) \to H^1(G, T(A_L))\Big),$$

induced by the embedding $T(L) \hookrightarrow T(A_L)$, where A_L is the ring of adeles.

1.3.2 Proposition (see [1], p. 147)**.**

$$Ш(T) = \mathrm{Ker}\Big(H^1(k, T(\bar{k})) \to \prod_v H^1(k_v, T(\bar{k_v}))\Big),$$

where $\bar{k}$ denotes an algebraic closure of k.

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2.1 Surjectivity of the Norm Map

2.1.1 Lemma. *Consider a cartesian square*

$$\begin{array}{ccc} L & \longrightarrow & L' \\ \uparrow & & \uparrow \\ k & \longrightarrow & k' \end{array}$$

Then

$$R^1_{L'/k'}(\mathbb{G}_m) \simeq R^1_{L/k}(\mathbb{G}_m) \underset{k}{\otimes} k'.$$

Proof. Fix $\{v_1, \cdots, v_n\}$ to be a basis of the extension L/k. Since the square is cartesian, the same basis can be taken for the extension L'/k'.

$$R^1_{L/k}(\mathbb{G}_m) \underset{k}{\times} k' = \mathrm{Spec}\Big(k[c_1, \cdots, c_n]/\big(\mathrm{N}(c_1v_1 + \cdots + c_nv_n) = 1\big)\Big) \underset{k}{\otimes} k' =$$
$$= \mathrm{Spec}\Big(k'[c_1, \cdots, c_n]/\big(\mathrm{N}(c_1v_1 + \cdots + c_nv_n) = 1\big)\Big) = R^1_{L'/k'}(\mathbb{G}_m).$$

□

2.1.2 Theorem. *Let $k = \mathbb{Q}$, let $L/\mathbb{Q}$ be an abelian Galois extension, let $T = R^1_{L/\mathbb{Q}}(\mathbb{G}_m)$ be the norm torus corresponding to this extension, and let $T_n = R^1_{L(\zeta_{p^n})/\mathbb{Q}(\zeta_{p^n})}(\mathbb{G}_m)$ be the torus obtained by base change in the tower of cyclotomic extensions (see 2.1.1):*

$$\mathbb{Q} \to \mathbb{Q}(\zeta_\ell) \to \mathbb{Q}(\zeta_{\ell^2}) \to \cdots \to \mathbb{Q}(\zeta_{\ell^\infty})$$

Assume also that the fields L and $\mathbb{Q}(\zeta_{\ell^k})$ are linearly disjoint for all k.

Then the norm map is surjective:

$$Ш(T_n) \twoheadrightarrow Ш(T_m), \qquad m \leq n.$$

Proof. Let $k_n = \mathbb{Q}(\zeta_{\ell^n})$ and $L_n = L(\zeta_{\ell^n})$.

In the long exact sequence of Tate cohomology associated with the short exact sequence

$$0 \to T(L) \to T(A_L) \to C_L(T) \to 0,$$

consider the fragment

$$\hat{H}^0(G, T(A_k)) \xrightarrow{\nu} \hat{H}^0(G, C_L(T)) \xrightarrow{\varphi} \hat{H}^1(G, T(k)) \xrightarrow{\psi} \hat{H}^1(G, T(A_L))$$

from which we obtain

$$Ш(T) = \mathrm{Ker}(\psi) = \mathrm{Coker}(\nu).$$

By the local and global Tate–Nakayama duality, see [1][pp. 144–145, Ths. 6.22–6.23],

$$\hat{H}^2(G_v, \hat{T}) \simeq \hat{H}^0(G_v, T(k_v))^\vee, \quad \hat{H}^2(G, \hat{T}) \simeq \hat{H}^0(G, C_L(T))^\vee.$$

Hence

$$Ш(T) = \mathrm{Ker}[H^2(G, \hat{T}) \to \prod_v H^2(G_v, \hat{T})]^\vee.$$

For the norm torus $T = R^1_{L/k}(\mathbb{G}_m)$, by definition,

$$\hat{T} = I_G^* = \mathrm{Hom}_{\mathbb{Z}[G]}(I_G, \mathbb{Z}),$$

where I_G is the augmentation ideal. Then from the sequence dual to the sequence defining I_G, we get

$$H^i(G, I_G^*) = H^{i+1}(G, \mathbb{Z}).$$

Substituting this into the formula above, we obtain

$$Ш(R^1_{L/k}(\mathbb{G}_m)) = \mathrm{Ker}[H^3(G, \mathbb{Z}) \to \prod_v H^3(G_v, \mathbb{Z})]^\vee.$$

It is known that

$$H^3(G, \mathbb{Z}) = \mathrm{Hom}(\wedge^2 G, \mathbb{Q}/\mathbb{Z})$$

for finite abelian groups, see [2][p. 123, Th. 6.4]. Therefore,

$$Ш(R^1_{L/k}(\mathbb{G}_m)) = \mathrm{Coker}\Big(\prod_{v \nmid \infty} \wedge^2 G_v \to \wedge^2 G\Big).$$

Under base change, the Galois groups are isomorphic:

$$\mathrm{Gal}(L_{n+1}/k_{n+1}) \simeq \mathrm{Gal}(L_n/k_n),$$

whereas the decomposition groups change as follows:

$$\begin{array}{ccccc} & & L_{n,v} & \longrightarrow & L_{n+1,v'} \\ & & \uparrow & & \uparrow \\ k_{n,w} & \longrightarrow & L_{n,v} \cap k_{n+1,w'} & \longrightarrow & k_{n+1,w'}. \end{array}$$

The right square is cartesian, whence

$$G_{n,v} = \mathrm{Gal}(L_{n,v}/k_{n,w}) \hookleftarrow \mathrm{Gal}(k_{n+1,w'} \cap L_{n,v}) \simeq \mathrm{Gal}(L_{n+1,v'}/k_{n+1,w'}) = G_{n+1,v'}.$$

Now consider the diagram

$$\begin{array}{ccc} Ш_{n+1} & \longrightarrow & Ш_n \\ \uparrow & & \uparrow \\ \wedge^2 G_{n+1} & \xrightarrow{\sim} & \wedge^2 G_n \\ \uparrow & & \uparrow \\ \prod \wedge^2 G_{n+1,v} & \longrightarrow & \prod \wedge^2 G_{n,v} \end{array}$$

Since the lower square is commutative, one can complete the arrow between the Ш, and the isomorphism

$$\wedge^2 G_{n+1} \simeq \wedge^2 G_n$$

yields a surjection on cokernels:

$$Ш(T_{n+1}) \twoheadrightarrow Ш(T_n).$$

□

2.2 The Weak Approximation Group

2.2.1 Definition. *The **weak approximation group** of a k-torus T is defined by*

$$A(T) = \Big(\prod_v T(k_v)\Big)/\overline{T(k)}.$$

We say that T satisfies weak approximation if $A(T) = 0$.

2.2.2 Theorem (see ([1], p. 152))**.** *Let L/k be a Galois extension of global fields with Galois group G, and let T be a k-torus split over L.*

Then, if all decomposition groups are cyclic, one has $A(T) = 0$.

2.3 Voskresenskii's Exact Sequence and Boundedness

2.3.1 Theorem (see [1], p. 154)**.** *Let L/k be an extension of global fields, and let T be the norm torus corresponding to this extension. Then there exists an exact sequence*

$$0 \to A(T) \to H^3(\mathrm{Gal}(L/k), \mathbb{Z})^\vee \to Ш(T) \to 0.$$

2.3.2 Corollary. *The groups $Ш(T_n)$ and $A(T_n)$ are uniformly bounded by $H^3(\mathrm{Gal}(L/k), \mathbb{Z})$.*

2.3.3 Corollary. *If the torus T splits over a cyclic extension, then*

$$Ш(T) = 0 \text{ and } A(T) = 0.$$

Proof. By 2-periodicity of the cohomology of cyclic groups,

$$H^3(\mathrm{Gal}(L/k), \mathbb{Z}) \simeq H^1(\mathrm{Gal}(L/k), \mathbb{Z}) = \mathrm{Hom}(\mathrm{Gal}(L/k), \mathbb{Z}) = 0,$$

since $\mathrm{Gal}(L/k)$ is finite. □

Let us fix the notation

$$T = R^1_{L/k}(\mathbb{G}_m), \qquad T_n = R^1_{L(\zeta_{p^n})/k(\zeta_{p^n})}(\mathbb{G}_m).$$

The behavior of the groups $Ш(T)$ and $A(T)$ depends on the arithmetic of the extension L/k.

2.3.4 Example. *Consider $L = \mathbb{Q}(\sqrt{13}, \sqrt{17})$ and $k = \mathbb{Q}$. In this case, since all decomposition groups are cyclic, it follows from Theorem (2.2.2) that $A(T) = 0$. Then Theorem (2.3.1) implies that $Ш(T) \simeq H^3(G, \mathbb{Z})$, and by the Künneth formula one has $H^3(G, \mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z}$. Therefore $Ш(T) \simeq \mathbb{Z}/2\mathbb{Z}$.*

Since the norm map surjects from $Ш(T_n)$ onto $Ш(T)$, we obtain

$$Ш(T) = Ш(T_n) = \mathbb{Z}/2\mathbb{Z},\ A(T) = A(T_n) = 0.$$

2.3.5 Example. *Consider $L = \mathbb{Q}(\sqrt{5}, \sqrt{13})$ and $k = \mathbb{Q}$.*

In this case, using the weak approximation criterion for norm tori, one can describe the behavior of the weak approximation group:

$$A(T) = \mathbb{Z}/2\mathbb{Z},\ A(T_1) = A(T_n) = 0.$$

Indeed, in the first case not all decomposition groups are cyclic, and the computation yields a nontrivial group, whereas in the second case all of them are cyclic.

The middle group in Voskresenskii's exact sequence remains unchanged; consequently, the Shafarevich–Tate group stabilizes and is equal to $\mathbb{Z}/2\mathbb{Z}$:

$$Ш(T) = 0,\ Ш(T_1) = Ш(T_n) = \mathbb{Z}/2\mathbb{Z}.$$

3 The Class Group of a Torus

3.1 Definitions and Examples

3.1.1 Definition. *Define the class group $Cl(T)$ of a k-torus T as the quotient*

$$T(A_k)/T(k)T(A_k^{S_\infty}),$$

where A_k is the ring of adeles, and $A_k^{S_\infty}$ is the open subring corresponding to the set of all archimedean places.

This quotient is compact and discrete, hence finite.
Denote its order by h_T.

3.1.2 Example. *From the adelic description of the class group of a number field k, it follows that*

$$Cl(\mathbb{G}_{m,k}) = I_k/k^\times I_k^{S_\infty} = Cl(k),$$

where I_k is the idèle group of k.

3.1.3 Example. *Similarly, for the torus obtained by Weil restriction of $\mathbb{G}_{m,L}$,*

$$Cl(R(\mathbb{G}_{m,L})) = I_L/L^\times I_L^{S_\infty} \simeq Cl(L).$$

3.2 The Case of Norm Tori

Consider the case of norm tori $T = R^1_{L/k}$ corresponding to a Galois extension L/k of global fields with Galois group G. In this case, the orders of the class groups of tori are computed by formulas obtained by Ono.

3.2.1 Theorem ([3], Th. 2.1)**.**

$$\frac{h_L}{h_T h_k} = \frac{|Ш(T)| \prod_v H^0(G_w, \mathcal{O}_w^\times)}{[L' : k] H^0(G, \mathcal{O}_L)},$$

where L' is the maximal abelian subextension of L/k, and in the product one chooses one w above each v.

3.3 Norm Tori of Imaginary Quadratic Extensions

Let us consider the simplest norm tori, namely those corresponding to imaginary quadratic extensions $\mathbb{Q}(\sqrt{-m})/\mathbb{Q}$. Let ℓ be a prime for which we consider the $\mathbb{Z}_\ell$-extension of $\mathbb{Q}(\sqrt{-m})$.

In this case, the formulas simplify to

$$h(T_n) := h(R^1_{L_n/k_n}) = \frac{h(L_n)}{h(k_n) 2^{\tau(L_n/k_n) - \varepsilon(L_n/k_n) - 1}},$$

where $\tau(n) = \tau(L_n/k_n)$ is the number of places that ramify, and $\varepsilon(n) = \varepsilon(L_n/k_n) = v_2\Big(\hat{H}^0(G_n, \mathcal{O}_{L_n}^\times)\Big)$.

4 Computations

```
Quadclassnormtori(l,m,n)=
{
local(kn,Lnkn,Ln,UULn,M,epsn,taun);
kn=bnfinit(polcyclo(l^n,y));
Lnkn=rnfinit(kn,x^2+m,1);
Ln=bnfinit(Lnkn[11][1]);
my(hLn = Ln.no);
my(hkn = kn.no);

UULn=bnfunits(Ln);
M=Mat([]);
for(i=1,Ln.r1+Ln.r2,
M=concat(M,bnfisunit(kn,rnfeltnorm(Lnkn,nffactorback(Ln,UULn[1][i])))));
M*Mod(1,2);
epsn=kn.r2-matrank(M*Mod(1,2));

taun=#idealfactor(kn,quaddisc(-m))[1,];

my(hn  = hLn/(hkn*2^(taun-epsn-1)));
```

```
return([["eps(n)","tau(n)","h(Ln)","h(kn)","h(n)"]; [epsn,taun,hLn,hkn,hn]]);
}
```

m	i	$\varepsilon(n)$	$\tau(n)$	$h(L_n)$	$h(k_n)$	$h(T_n)$
1	1	0	1	1	1	1
1	2	0	1	1	1	1
2	1	0	1	1	1	1
2	2	0	1	3	1	3
5	1	1	2	2	1	2
5	2	1	2	6	1	6
7	1	1	2	1	1	1
7	2	1	2	1	1	1
10	1	1	2	2	1	2
10	2	1	2	14	1	14
11	1	0	1	1	1	1
11	2	0	1	3	1	3
13	1	1	3	4	1	2
13	2	1	3	52	1	26
14	1	1	3	4	1	2
14	2	1	3	36	1	18
17	1	1	2	4	1	4
17	2	3	4	48	1	48
19	1	1	2	1	1	1
19	2	3	6	4	1	1
22	1	1	2	2	1	2
22	2	1	2	38	1	38
23	1	0	1	3	1	3
23	2	0	1	9	1	9
26	1	0	3	12	1	3
26	2	0	3	252	1	63
29	1	1	2	6	1	6
29	2	1	2	126	1	126
31	1	1	2	3	1	3
31	2	1	2	9	1	9
34	1	1	2	4	1	4
34	2	3	4	112	1	112
35	1	1	3	2	1	1
35	2	1	3	18	1	9
37	1	1	3	4	1	2
37	2	3	7	208	1	26
38	1	1	3	6	1	3
38	2	3	7	288	1	36
41	1	1	2	8	1	8
41	2	1	2	216	1	216
43	1	1	2	1	1	1
43	2	1	2	13	1	13
46	1	1	2	4	1	4
46	2	1	2	172	1	172
47	1	0	1	5	1	5
47	2	0	1	45	1	45

References

[1] V.E. Voskresenskii, *Algebraicheskie tory*, Nauka, 1977.

[2] K.S. Brown, *Cohomology of Groups*, Springer, 1982.

[3] T. Ono, *On some class number relations for Galois extensions*, Nagoya Math. J., 1987.